Member 1

Name: Subhana Nawar Sadita **Id: 23101004**

Topology Design and Physical Setup

- Designed the overall network topology for all 6 units
- Added all routers, switches, PCs, printers and servers in Cisco Packet Tracer
- Connected all devices using correct cables (Serial DCE and Copper Straight-Through)
- Added HWIC-2T serial modules to all routers
- Configured IP addresses on HO-Router and TC-Router interfaces
- Configured DHCP on TC-Router for the Training Center network

Member 2

Name: Muaz Mohammad Zimam **Id: 23101529**

IP Addressing, VLSM and DHCP

- Calculated the base network address from the group student ID (13.13.0.0/16)
- Performed VLSM subnetting for all 6 units in correct order
- Assigned all inter-router link addresses using /30 subnets
- Configured IP addresses on SL-Router and SD-Router interfaces
- Configured dedicated DHCP servers for SL and SD units
- Assigned static IP addresses to all HO and AU devices

Member 3

Name: Aysha Siddiqua Tithi **Id: 22301667**

Routing Configuration (Static and RIPv2)

- Configured IP addresses on RB-Router and AU-Router interfaces
- Configured DHCP on RB-Router for the Remote Branch network
- Implemented all required static routing types:
    1. Next-hop static routing on HO-Router toward TC
    2. Default static routing on TC-Router toward HO
    3. Recursive static routing on SD-Router toward TC via SL
    4. Exit-interface static routing on RB-Router toward HO
    5. Floating static routing on AU-Router toward HO via RB (AD 200)
- Configured RIPv2 on HO, SL, SD, RB and AU routers
- Configured route redistribution of static routes into RIPv2 on HO-Router

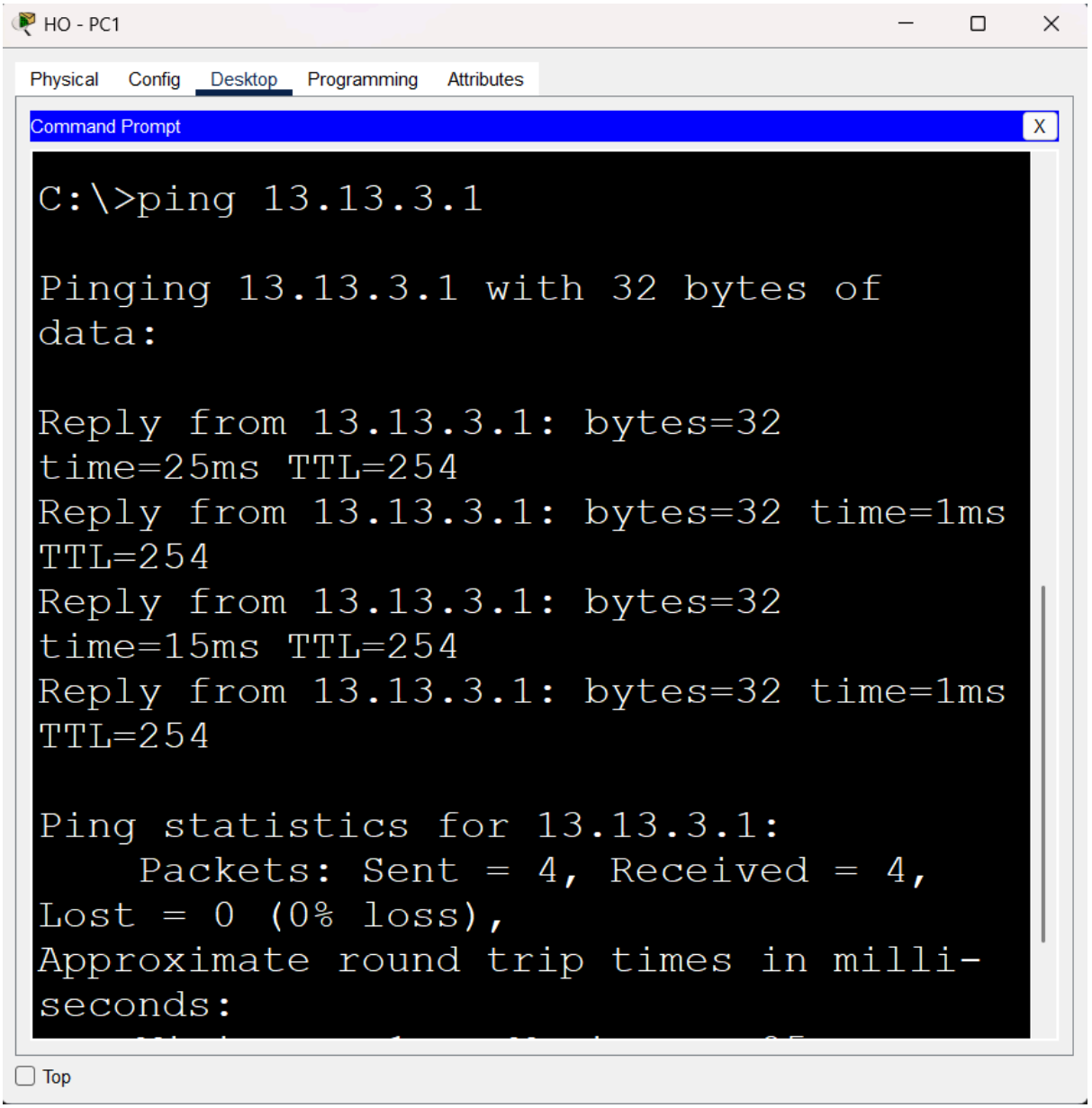
Member 4

Name: M A Obaeed **Id: 21201382**

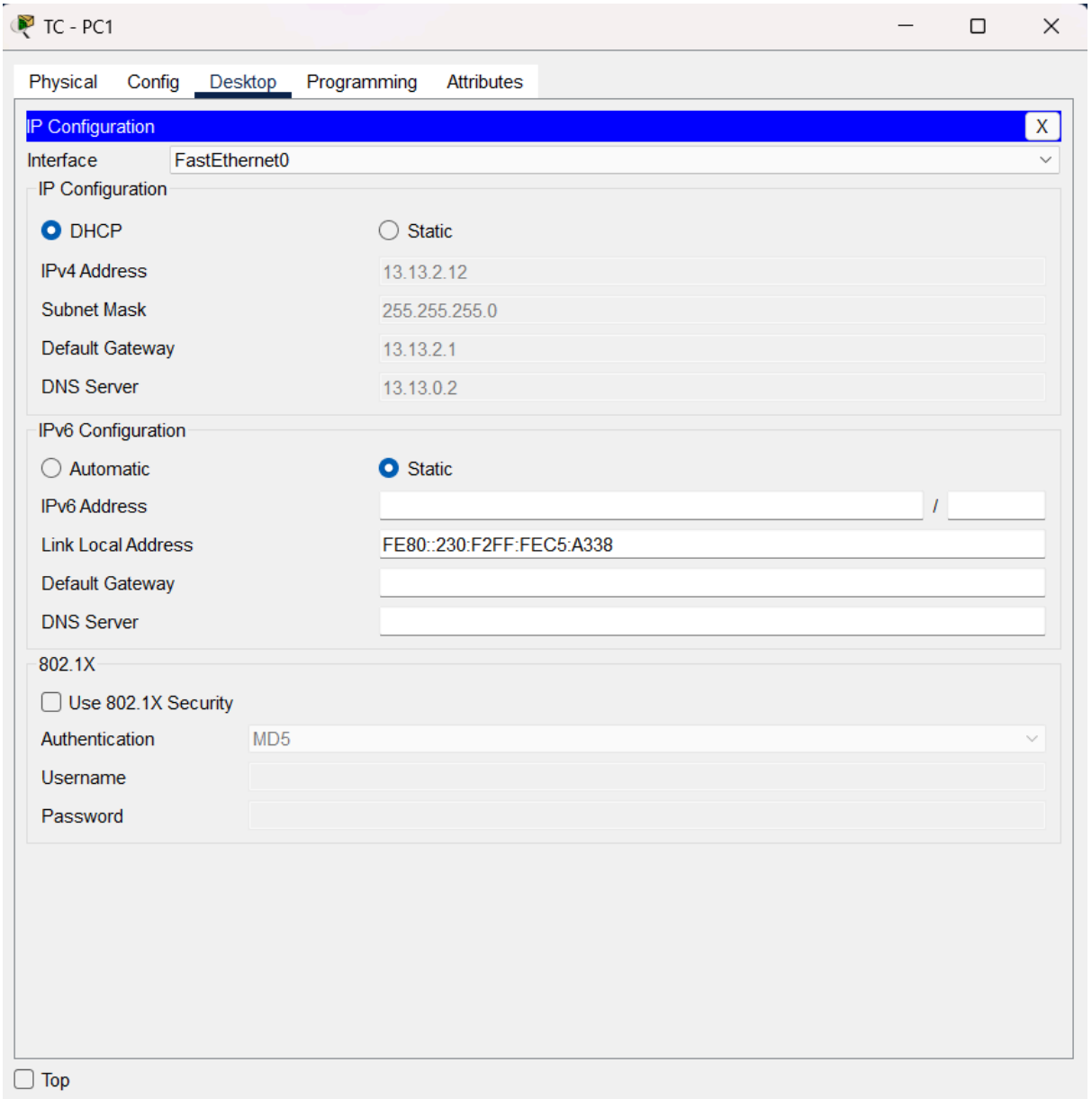
Server Configuration and Testing

- Configured DNS server on HO-DNSWebServer with www.sisn.local record
- Configured HTTP Web server with SISN welcome page
- Configured email servers on all 6 units with correct domains
- Added DNS A records for all email server domains
- Configured email clients on HO-PC1 and SL-PC1
- Tested and verified email communication between HO and SL
- Performed full network connectivity testing including ping tests
- Tested fault tolerance by simulating link failures
- Verified floating static route activation when primary path fails
- Took screenshots of all test results for submission

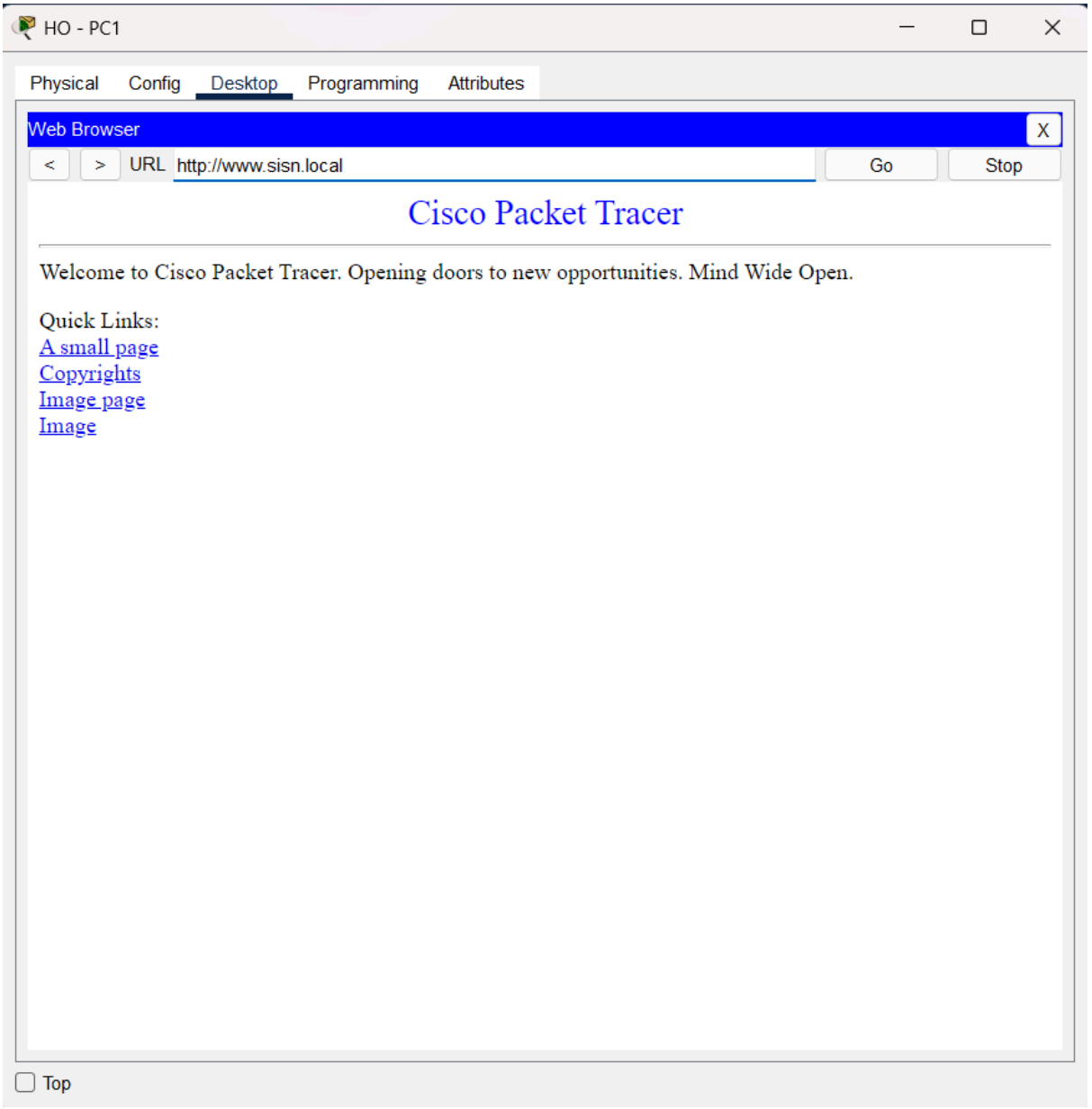
Proof 1 — Ping Test



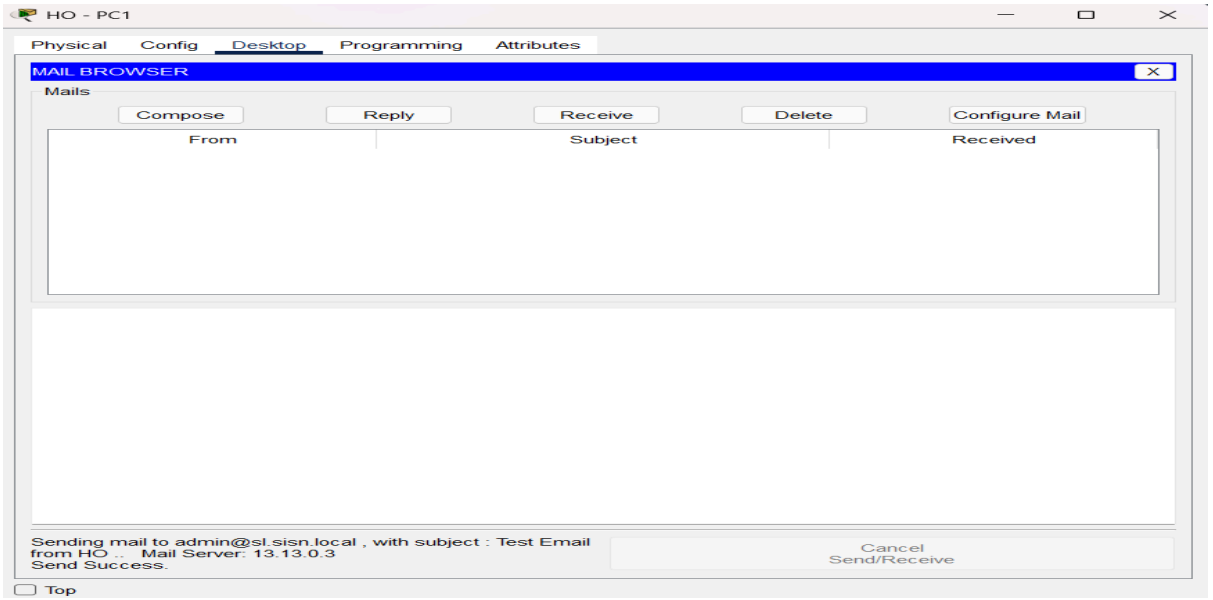
Proof 2 — DHCP Working

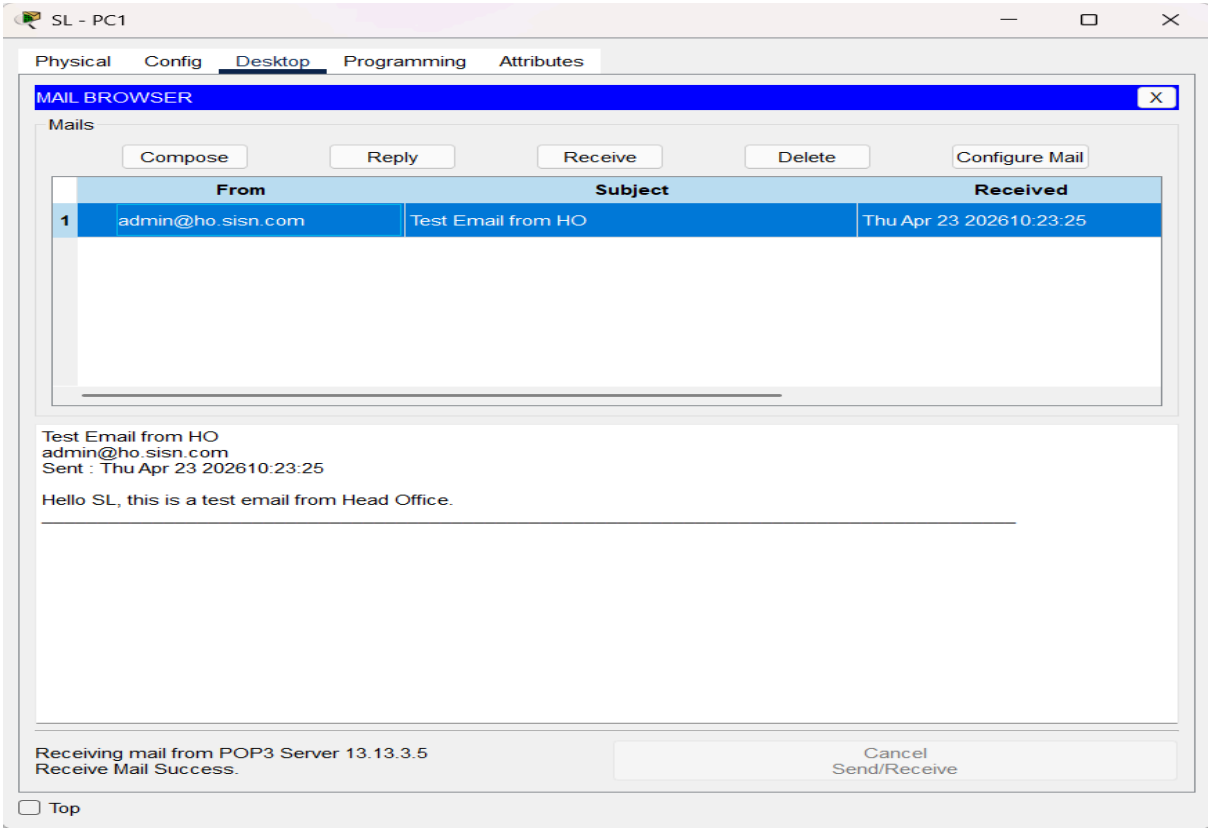


Proof 3 — DNS and Web Server Working



Proof 4 — Email Communication Working





Proof 5 — Routing Table Proof

